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Apparatus and method for providing MBS data to user equipment in cellular communication system

Abstract

A method for Xn handover of a multicast/broadcast service (MBS) by a session management function (SMF) device in a mobile communication system is provided. The method includes receiving, from a target next generation radio access network (NG-RAN) node through an access and mobility management function (AMF), a first message including information on whether the target NG-RAN node supports the MBS, determining an individual delivery method for MBS data, in case that the target NG-RAN node does not support the MBS based on the first message, and setting up the MBS data to be individually delivered to a user equipment (UE) through the target NG-RAN node.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2019/0158985	12/2018	Dao	N/A	H04W 28/04
2021/0168568	12/2020	Dao	N/A	H04L 12/185
2021/0352444	12/2020	Griot	N/A	H04W 8/02
2022/0338069	12/2021	Wang	N/A	H04W 76/27
2023/0065481	12/2022	Xiong	N/A	H04W 36/0011
2023/0188949	12/2022	Jia	455/458	H04L 12/1881
2023/0292227	12/2022	Wang	N/A	H04W 36/0007
2023/0309189	12/2022	R?nneke	N/A	H04W 4/06
2023/0345310	12/2022	Li	N/A	H04W 76/40

OTHER PUBLICATIONS

International Search Report dated Jun. 23, 2022, issued an International Application No. PCT/KR2022/004611. cited by applicant

Nokia et al., ‘Mobility from MBS Supporting to Non-Supporting MBS nodes’, R3-210173, 3GPP TSG RAN WG3 #111-e; Jan. 14, 2021. cited by applicant

CMCC, ‘MBS Session management over NG’, R3-210920, 3GPP TSG RAN WG3 #111-e; Jan. 15, 2021. cited by applicant

Huawei et al., ‘Mobility Procedures for 5MBS’, S2-2100718, 3GPP TSG SA2 Meeting #143-e; Feb. 18, 2021. cited by applicant

Ericsson, ‘On Support of mobility between gNBs supporting MBS’, R3-210643, 3GPP TSG RAN WG3 #111-e; Jan. 15, 2021. cited by applicant

CATT: “Mobility support of MBS service”, 3GPP Draft; S2-2100590, 3rd Generation Partnership Project (3GPP), Mobile Competence Centre; 650, Route Des Lucioles; F-06921 Sophia-Antipolis Cedex; France, vol. SA WG2, No. E (e-meeting); Feb. 24, 2020-Mar. 9, 2020, XP052172921, Feb. 18, 2021. cited by applicant

ZTE: “KI#7, new solution multicast MBS session handover with Dummy QoS Flow”, 3GPP Draft; S2-2004509, 3rd Generation Partnership Project (3GPP), Mobile Competence Centre; 650, Route Des Lucioles; F-06921 Sophia-Antipolis Cedex; France, vol. SA WG2, No. Electronic, Elbonia; Jun. 1, 2020-Jun. 12, 2020, XP052461217, Jun. 12, 2020. cited by applicant

Shanghai Jiao Tong University: “Reliable delivery mode switching within single RAN/dual RANs”, 3GPP Draft; S2-2006325, 3rd Generation Partnership Project (3GPP), Mobile Competence Centre; 650, Route Des Lucioles; F-06921 Sophia-Antipolis Cedex; France, vol. SA WG2, No. Online; Aug. 19, 2020-Sep. 2, 2020, XP052463738, Sep. 2, 2020. cited by applicant

Extended European Search Report dated Jun. 24, 2024, issued in European Application No. 22781651.9. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application is based on and claims priority under 35 U.S.C. § 119(a) of a Korean patent application number 10-2021-0042380, filed on Mar. 31, 2021, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

(2) The disclosure relates to a method and apparatus for transmitting multicast and/or broadcast data to a terminal. More particularly, the disclosure relates to a method and apparatus for transmitting multicast and/or broadcast data from a 5th-generation (5G) network to a terminal.

2. Description of Related Art

(3) Considering the development of wireless communication from generation to generation, the technologies have been developed mainly for services targeting humans, such as voice calls, multimedia services, and data services. Following the commercialization of 5th-generation (5G) communication systems, it is expected that the number of connected devices will exponentially grow. Increasingly, these will be connected to communication networks. Examples of connected things may include vehicles, robots, drones, home appliances, displays, smart sensors connected to various infrastructures, construction machines, and factory equipment. Mobile devices are expected to evolve in various form-factors, such as augmented reality glasses, virtual reality headsets, and hologram devices. In order to provide various services by connecting hundreds of billions of devices and things in the 6th-generation (6G) era, there have been ongoing efforts to develop improved 6G communication systems. For these reasons, 6G communication systems are referred to as beyond-5G systems.

(4) 6G communication systems, which are expected to be commercialized around 2030, will have a peak data rate of tera (1,000 giga)-level bps and a radio latency less than 100 μsec, and thus will be 50 times as fast as 5G communication systems and have the 1/10 radio latency thereof.

(5) In order to accomplish such a high data rate and an ultra-low latency, it has been considered to implement 6G communication systems in a terahertz band (for example, 95 GHz to 3 THz bands). It is expected that, due to severer path loss and atmospheric absorption in the terahertz bands than those in mmWave bands introduced in 5G, technologies capable of securing the signal transmission distance (that is, coverage) will become more crucial. It is necessary to develop, as major technologies for securing the coverage, radio frequency (RF) elements, antennas, novel waveforms having a better coverage than orthogonal frequency division multiplexing (OFDM), beamforming and massive multiple input multiple output (MIMO), full dimensional MIMO (FD-MIMO), array antennas, and multiantenna transmission technologies such as large-scale antennas. In addition, there has been ongoing discussion on new technologies for improving the coverage of terahertz-band signals, such as metamaterial-based lenses and antennas, orbital angular momentum (OAM), and reconfigurable intelligent surface (RIS).

(6) Moreover, in order to improve the spectral efficiency and the overall network performances, the following technologies have been developed for 6G communication systems: a full-duplex technology for enabling an uplink transmission and a downlink transmission to simultaneously use the same frequency resource at the same time; a network technology for utilizing satellites, high-altitude platform stations (HAPS), and the like in an integrated manner; an improved network

structure for supporting mobile base stations and the like and enabling network operation optimization and automation and the like; a dynamic spectrum sharing technology via collision avoidance based on a prediction of spectrum usage; an use of artificial intelligence (AI) in wireless communication for improvement of overall network operation by utilizing AI from a designing phase for developing 6G and internalizing end-to-end AI support functions; and a next-generation distributed computing technology for overcoming the limit of UE computing ability through reachable super-high-performance communication and computing resources (such as mobile edge computing (MEC), clouds, and the like) over the network. In addition, through designing new protocols to be used in 6G communication systems, developing mechanisms for implementing a hardware-based security environment and safe use of data, and developing technologies for maintaining privacy, attempts to strengthen the connectivity between devices, optimize the network, promote softwarization of network entities, and increase the openness of wireless communications are continuing.

(7) It is expected that research and development of 6G communication systems in hyper-connectivity, including person to machine (P2M) as well as machine to machine (M2M), will allow the next hyper-connected experience. Particularly, it is expected that services such as truly immersive extended reality (XR), high-fidelity mobile hologram, and digital replica could be provided through 6G communication systems. In addition, services such as remote surgery for security and reliability enhancement, industrial automation, and emergency response will be provided through the 6G communication system such that the technologies could be applied in various fields such as industry, medical care, automobiles, and home appliances.

(8) In order to transmit the same data to a plurality of terminals positioned in a specific region in a mobile communication network, data may be transmitted to each terminal, for example, a user equipment by unicast. Further, in some cases, for resource efficiency, it is necessary to provide a data service to a plurality of terminals in a mobile communication network through multicast/broadcast.

(9) In this case, there is a need for a method of supporting a smooth reception of a service even when a terminal receiving a multicast or broadcast service moves and a base station thus changes.

(10) Such a multicast service has been provided since the early 2000s, and as cellular networks develop, it is necessary to provide a multicast service in a manner that meets the standard of each cellular network. However, a method for appropriately providing a multicast service has not yet been provided in the 5G network.

(11) The above information is presented as background information only to assist with an understanding of the disclosure. No determination has been made, and no assertion is made, as to whether any of the above might be applicable as prior art with regard to the disclosure.

SUMMARY

(12) In order to provide a multicast service, a 5th-Generation system (5GS) receives multicast service data from an application function (AF) or a content provider and delivers the multicast service data to a next generation radio access network (NG-RAN), which is a base station, thus, the 5GS may transmit multicast service data to terminals, for example, user equipments (UEs) subscribing to the multicast service. There are two methods for a 5G core network to deliver multicast data to the NG-RAN shared delivery and individual delivery. When the NG-RAN has a multicast/broadcast service (MBS) capability, multicast service data may be transmitted from a multicast/broadcast user plane function (MB-UPF) device to the NG-RAN through a tunnel for shared delivery. However, because shared delivery is impossible when the NG-RAN does not have an MBS capability, it is possible to transmit multicast service data such as MBS data received through an MB-UPF as individual delivery is transmitted to the terminal through an associated protocol data unit (PDU) session and through a tunnel from the UPF to the NG-RAN. Therefore, as the terminal moves, there is a need for a method of smoothly supporting a multicast service to the mobile terminal according to whether the target NG-RAN supports an MBS capability.

(13) Aspects of the disclosure are to address at least the above-mentioned problems and/or disadvantages and to provide at least the advantages described below. Accordingly, an aspect of the disclosure is to provide a method and apparatus for transmitting multicast and/or broadcast data from a 5G network to a terminal.
(14) Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.
(15) In accordance with an aspect of the disclosure, a method for Xn handover of a multicast/broadcast service (MBS) by a session management function (SMF) device in a mobile communication system is provided. The method includes receiving, from a target next generation radio access network (NG-RAN) node through an access and mobility management function (AMF), a first message including information on whether the target NG-RAN node supports the MBS, determining an individual delivery method for MBS data, in case that the target NG-RAN node does not support the MBS based on the first message, and setting up the MBS data to be individually delivered to a user equipment (UE) through the target NG-RAN node.
(16) In accordance with another aspect of the disclosure, a session management function (SMF) for Xn handover of a multicast/broadcast service (MBS) in a mobile communication system is provided. The SMF includes a network interface configured to communicate with network functions (NFs) in the mobile communication system, a memory configured to store information, and at least one processor configured to control to receive, from a target next generation radio access network (NG-RAN) node through an access and mobility management function (AMF) using the network interface, a first message including information on whether the target NG-RAN node supports the MBS, determine an individual delivery method for MBS data, in case that the target NG-RAN node does not support the MBS based on the first message, and set up the MBS data to be individually delivered to a user equipment (UE) through the target NG-RAN node.
(17) Other aspects, advantages, and salient features of the disclosure will become apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses various embodiments of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:
(2) FIG. **1** illustrates a 5th-Generation System (5GS) structure for a multicast service according to an embodiment of the disclosure;
(3) FIG. **2** illustrates a process of supporting Xn handover to a User Equipment (UE) in a multicast service according to an embodiment of the disclosure;
(4) FIG. **3** illustrates a process of supporting Xn handover to a UE in a multicast service according to an embodiment of the disclosure;
(5) FIG. **4** illustrates a process of supporting Xn handover to a UE in a multicast service according to an embodiment of the disclosure;
(6) FIG. **5** illustrates a process of supporting Xn handover to a UE in a multicast service according to an embodiment of the disclosure;
(7) FIG. **6** illustrates a process of supporting N2 handover to a UE in a multicast service according to an embodiment of the disclosure;
(8) FIG. **7** illustrates a process of supporting N2 handover to a UE in a multicast service according to an embodiment of the disclosure;
(9) FIG. **8** illustrates a process of supporting N2 handover to a UE in a multicast service according

to an embodiment of the disclosure;
(10) FIG. **9** illustrates a process of supporting N2 handover to a UE in a multicast service according to an embodiment of the disclosure;
(11) FIG. **10** is a block diagram illustrating a configuration of a terminal to which the disclosure may be applied according to an embodiment of the disclosure; and
(12) FIG. **11** is a block diagram illustrating a configuration of a network function device to which the disclosure may be applied according to an embodiment of the disclosure.
(13) Throughout the drawings, it should be noted that like reference numbers are used to depict the same or similar elements, features, and structures.
DETAILED DESCRIPTION
(14) The following description with reference to the accompanying drawings is provided to assist in a comprehensive understanding of various embodiments of the disclosure as defined by the claims and their equivalents. It includes various specific details to assist in that understanding but these are to be regarded as merely exemplary. Accordingly, those of ordinary skill in the art will recognize that various changes and modifications of the various embodiments described herein can be made without departing from the scope and spirit of the disclosure. In addition, descriptions of well-known functions and constructions may be omitted for clarity and conciseness.
(15) The terms and words used in the following description and claims are not limited to the bibliographical meanings, but, are merely used by the inventor to enable a clear and consistent understanding of the disclosure. Accordingly, it should be apparent to those skilled in the art that the following description of various embodiments of the disclosure is provided for illustration purpose only and not for the purpose of limiting the disclosure as defined by the appended claims and their equivalents.
(16) It is to be understood that the singular forms "a," "an," and "the" include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to "a component surface" includes reference to one or more of such surfaces.
(17) Hereinafter, a term identifying an access node used in the description, a term indicating network entities, a term indicating messages, a term indicating an interface between network objects, a term indicating various types of identification information and the like are exemplified for convenience of description. Accordingly, the disclosure is not limited to the terms described below, and other terms indicating objects having equivalent technical meanings may be used.
(18) Hereinafter, for convenience of description, the disclosure uses terms and names defined in the standard for a 5G system. However, the disclosure is not limited by the above terms and names, and may be equally applied to systems conforming to other standards.
(19) FIG. **1** illustrates a structure of a cellular system for an MBS service according to an embodiment of the disclosure.
(20) Referring to FIG. **1**, the cellular system may include a user equipment (UE) **10**, an NG radio access network (NG-RAN) **20**, which is a base station, an access and mobility management function (AMF) device **101**, a multicast/broadcast user plane function (MB-UPF) device **111**, a multicast/broadcast-session management function (MB-SMF) device **112**, a policy control function (PCF) device **105**, a session management function (SMF) device **107**, a network exposure function (NEF) device **106**, a multicast/broadcast service function (MBSF) device **122**, a multicast/broadcast service transport function (MBSTF) device **121**, an application function (AF) device **130**, a unified data management (UDM) device **102**, a user plane function (UPF) device **108**, an authentication server function (AUSF) device **103**, and an NF repository function (NRF) device **104**.
(21) In describing FIG. **1**, each network function (NF) of the 5GS will be described as a "network function device" or a "network function" itself. However, those skilled in the art may know that the NF and/or the NF device may be implemented in one or more specific servers, and that two or more NFs performing the same operation may be implemented in one server.

(22) One NF or two or more NFs may be implemented in the form of one network slice in some cases. Such a network slice may be generated based on a specific purpose. For example, the network slice may be configured for a subscriber group for providing the same type of service, e.g., a maximum data rate, a data usage rate, and a guaranteed minimum data rate to specific subscriber groups. Further, the network slice may be implemented for various purposes, and a further description will be omitted herein.
(23) Further, each node interface is illustrated in FIG. **1**. A Uu interface is used between the UE **10** and the NG-RAN **20**, an N2 interface is used between the NG-RAN **20** and the AMF **101**, and an N3-shared interface is used between the NG-RAN **20** and the MB-UPF **111**. Further, an N4-MBS interface is used between the MB-UPF **111** and the MB-SMF **112**, and an N9-MBS interface is used between the MB-UPF **111** and the UPF **108**. An N4 interface is used between the SMF **107** and the UPF **108**, an N6 interface is used between the UPF **108** and the AF **130**, and an Nxxx interface is used between the MBSF **122** and the MBSTF **121**. Because these interfaces are defined in the NR standard, a further description will be omitted here.
(24) In general, in order to support the MBS service in the 5GS, a cellular system for the MBS may be configured with the following network function devices and services.
(25) The AF **130** may be implemented into, for example, a V2X application server, a CIoT application server, an MCPTT application, a contents provider, a TV or audio service provider, and a streaming video service provider.
(26) In order to provide an MBS service to a specific user or a specific user group, the AF **130** may request the provision of the MBS service to the MBSF **122**, which is an NF that controls session management and traffic of the MBS service. For example, the MBSF **122** may be an NF that receives a request for an MBS service from the AF to manage a corresponding MBS service session, and to control the corresponding MBS service traffic. Further, the MBSTF **121** is an NF that receives media from AF/AS or a contents provider based on the control of the MBSF **122** to process media traffic and may operate as an MBS service anchor in the 5GS. In the disclosure, the AF **130** may be an application server (AS) for providing a specific multicast/broadcast application service. Therefore, hereinafter, it may be understood that the AS is the same as the AF **130** or that the AF **130** and the AS exist together. In order to provide the MBS service, when the AF **130** transmits a request for providing the MBS service to the UE **10** to the MBSF **122**, the MBSF **122** may control the MBSTF **121**, which is an MBS service media anchor in the 5GS that transmits MBS service traffic to the UE **10** so that the MBS service is provided to the UE **10**. In this case, the MBS service is a service received from a specific contents provider, and may be one of various types of services exemplified above or other services.
(27) According to an embodiment, the MBSF **122** and the MBSTF **121** may be configured to be integrated into one entity or one NF. As another example, the MBSF **122** may be configured to be integrated into the NEF **106** or another NF. As another example, in the 5GS, the AF **130** may directly request the MBS service to the MB-SMF **112** without the MBSF **122** and the MBSTF **121**, and the MB-UPF **111** may receive media from a content provider, which is the AS or the AF **130** to forward traffic.
(28) The MBS service session is managed and service traffic is generated through the MBSF **122** and the MBSTF **121**, and when the service traffic is delivered to the UE **10** through multicast/broadcast, corresponding traffic may be managed by allocating an MBS PDU session. That is, the MBSF **122** may correspond to a control plane that manages the MBS session, and the MBSTF **121** may correspond to a user plane that handles traffic.
(29) In the following description, a “multicast-broadcast service gateway-control plane (MBMS-GW-C) service” is referred to as a control function or service for generating an MBS context for the MBS PDU session, managing the MBS PDU session and delivering traffic of the MBS PDU session to the NG-RAN **20**, which is a base station through IP multicast.
(30) The MBMS-GW-C service may be integrated into an existing SMF that manages a unicast

PDU session to be configured as an SMF with an MBS PDU session control function or may be configured as a separate NF. An NF supporting the MBMS-GW-C service and having a function of an existing SMF is referred to as an MB-SMF **112** in the disclosure.
(31) Further, a service that delivers traffic received from the MB-UPF **111** according to an MBS context for the MBS PDU session through IP multicast to the NG-RAN **20** that performs multicast/broadcast according to the MBMS-GW-C service is referred to as a multimedia broadcast-multicast service gateway-user plane (MBMS-GW-U) service.
(32) The MBMS-GW-U service may be configured with a UPF with a function of being integrated into the existing UPF that processes a unicast PDU session to deliver MBS traffic to an appropriate NG-RAN by IP multicast or may be configured with a separate NF, as illustrated in FIG. **1**. Therefore, in the following description, an NF that supports the MBMS-GW-U service and that has together a function of the existing UPF will be referred to as the MB-UPF **111**.
(33) As described above, the MBMS-GW-C service uses an N4-MBS interface so as to control the MBMS-GW-U service.
(34) In describing various embodiments of the disclosure, MBMS-GW-C and MBMS-GW-U are mainly described with the names SMF and UPF or the MB-SMF **112** and the MB-UPF **111**, respectively, for convenience, but whether the use is unicast-only, multicast/broadcast-only, or both is described together to avoid confusion, as needed.
(35) MBS traffic is delivered from the MBMS-GW-U (or UPF or MB-UPF) to the NG-RANs **20**. For example, MBS traffic is delivered to the NG-RAN **20** using IP multicast. In this case, a tunnel between the MBMS-GW-U (or UPF or MB-UPF) and the NG-RAN is referred to as an M1 tunnel, a shared delivery tunnel, or a shared N3 tunnel.
(36) In order to establish the M1 tunnel, the MBMS-GW-C (or SMF or MB-SMF) may transmit a control message to the NG-RAN **20** through the AMF.
(37) FIG. **2** illustrates a process of supporting Xn handover to a UE in a multicast service according to an embodiment of the disclosure.
(38) In the following description, a serving NG-RAN, which is a serving base station, will be described with reference numeral **21**, and a target NG-RAN, which is the target for handover, will be described with reference numeral **22**. Further, in the disclosure, there may be a plurality of SMFs, and accordingly, they are indicated as SMF(s). However, it should be noted that reference numeral **107** illustrated in FIG. **1** is used for the SMF for convenience of description. Further, a plurality of UPFs may be used in the disclosure. It is noted that reference numeral **108** illustrated in FIG. **1** is used for the UPF for convenience of description.
(39) Referring to FIG. **2**, in operation **200**, handover preparation (HO preparation) is performed between a serving NG-RAN **21** and a target NG-RAN **22**, and in operation **201**, Xn handover may be performed through HO execution. That is, the UE **10** may perform a handover procedure from the serving NG-RAN **21**, which is a base station that was initially provided with a service to the target NG-RAN **22**.
(40) The target NG-RAN **22** may transmit an N2 path switch request message to the AMF **101** in operation **202** in order to move a data path flowing to the existing serving NG-RAN **21** to itself. In this case, the N2 path switch request message may be transmitted to the AMF **101** by including the N2 session management (SM) message sending to each SMF **107** or MB-SMF **112** for each session. In this case, the target NG-RAN **22** may include an MBS capability indicating whether it supports 5 MBS in the N2 path switch request.
(41) The SM message may include tunnel endpoint information (address information of the target NG-RAN, tunnel ID, and the like) for the target NG-RAN **22** to receive data traffic for the corresponding session.
(42) Upon receiving the N2 path switch request message in operation **202**, the AMF **101** may transmit an Nsmf_PDUSession_UpdateSMcontext request message to the corresponding SMFs **107** in order to deliver the SM message sending to the SMF **107** in operation **203**. The

Nsmf_PDUSession_UpdateSMcontext request may include a 5 MBS capability of the target NG-RAN **22** from capability information of the NG-RAN already known by the AMF **101** through a configuration procedure, and the like separately from the N2 SM message.

(43) In operation **204**, the SMF **107** may determine to use an individual delivery method as a data traffic delivery method corresponding to the MBS session identifier (Session ID) according to the received 5 MBS capability of the target NG-RAN **22**. The SMF **107** may know a 5 MBS capability through the capability transmitted by the target NG-RAN **22** with a method of identifying the 5 MBS capability of the target NG-RAN **22**. The SMF **107** may know a 5 MBS capability by distinguishing whether an assigned tunnel ID in tunnel endpoint information sent by the target NG-RAN **22** is an ID for an individual tunnel or an ID for a shared tunnel with another method of identifying a 5 MBS capability of the target NG-RAN **22**. That is, an ID range for the shared tunnel and an ID range for the individual tunnel may be operated separately or an information element (IE) for ID delivery for a shared tunnel may be distinguished by using an IE separately from an ID for the individual tunnel.

(44) Therefore, when a 5 MBS capability of the target NG-RAN **22** is not supported, it is determined to individual delivery, and when a 5 MBS capability of the target NG-RAN **22** is supported, it is determined to shared delivery.

(45) Operations **211**, **212** and **213** are shown in **210** in FIG. **2**. Accordingly, in operation **211**, the SMF **107** may transmit an Nsmf_MBSSession_UpdateSMcontext request message to the MB-SMF **112** by including the corresponding MBS session identifier (ID). Alternatively, the Nsmf_MBSSession_UpdateSMcontext request may include tunnel endpoint information of a PDU session anchor user plane function (PSA-UPF), for example, an address of the UPF and a tunnel identifier (tunnel ID) of the UPF for receiving MBS traffic corresponding to the MBS session ID. In this case, the SMF **107** may include an associated PDU session ID connected with the MBS session ID in the Nsmf_MBSSession_UpdateSMcontext request message.

(46) The MB-SMF **112** that has received the information may transmit the received tunnel endpoint information of the PSA-UPF to the MB-UPF **111** in order to deliver traffic corresponding to the MBS session ID to the PSA-UPF in operation **212**. The MB-UPF **111** may transmit a response message including tunnel endpoint information (e.g., MB-UPF address, tunnel ID, lower layer MAC address, and the like) thereof to the MB-SMF **112** in response thereto. In operation **213**, the MB-SMF **112** may transmit the information included in the response message of operation **212** to the SMF **107** in response to the request in operation **211**.

(47) Accordingly, the SMF **107** that has received the tunnel endpoint information of the MB-UPF **111** may update the PDU session ID connected with the MBS session ID for individual delivery. Further, for this purpose, in operation **221**, the SMF **107** may generate an individual tunnel between the PSA-UPF and the MB-UPF **111** for the connected PDU session ID by delivering to the UPF **108** operating as the PSA-UPF. Further, the SMF **107** may transmit tunnel endpoint information of the target NG-RAN received through the AMF **101** in operation **203** to the UPF **108** in operation **221** in order to penetrate to generate an individual tunnel between the target NG-RAN **22** and the UPF **108** operating as a PSA-UPF.

(48) The SMF **107** may transmit the tunnel endpoint information of the target NG-RAN **22** received in operation **203** to the UPF **108**; thus, N4 session modification setting up an individual tunnel with the target NG-RAN **22** may be performed separately after operation **204**. Accordingly, the UPF **108** may generate an N4 session modification response message and transmit the N4 session modification response message to the SMF **107** in operation **222**.

(49) Because Xn handover is completed through operations **223** to **226**, when the target NG-RAN **22** does not have an MBS capability, the target NG-RAN **22** may provide smoothly a multicast service to the UE **10** through individual delivery.

(50) When describing this in more detail, the UPF **108** may generate an N3 end marker message to transmit the N3 end marker message to the serving NG-RAN **21** in operation **223**, thereby

notifying the target NG-RAN **22** of the change of the path. Further, such an N3 End marker message is delivered from the serving NG-RAN **21** to the target NG-RAN **22**, so that both the serving NG-RAN **21** and the target NG-RAN **22** may recognize that the path has been changed. Further, the SMF **107** may generate an Nsmf_PDUSession_UpdateSMcontext response message in response to operation **203** and transmit the Nsmf_PDUSession_UpdateSMcontext response message to the AMF **101** in operation **224**. Accordingly, the AMF **101** may generate and transmit an N2 path switch request ACK message to the target NG-RAN **22** in operation **225**, thereby notifying that N2 path switching has been performed. Thereafter, the target NG-RAN **22** may transmit a release resource message to the serving NG-RAN **21** in operation **226**, thereby completing the handover procedure. Accordingly, the path of MBS traffic to be provided to the UE **10** connected to the target NG-RAN **22** is changed from the serving NG-RAN **21** to the target NG-RAN **22**.

(51) FIG. **3** illustrates a process of supporting Xn handover to a UE in a multicast service according to an embodiment of the disclosure.

(52) As described above with reference to FIG. **2**, in the description, a serving NG-RAN, which is a serving base station, will be described with reference numeral **21**, and a target NG-RAN, which is the target for handover, will be described with reference numeral **22**. Further, in the disclosure, there may be a plurality of SMFs, and accordingly, they are indicated as SMF(s). However, it should be noted that reference numeral **107** illustrated in FIG. **1** is used for the SMF for convenience of description. Further, a plurality of UPFs may be used in the disclosure. It is noted that reference numeral **108** illustrated in FIG. **1** is used for the UPF for convenience of description.

(53) Referring to FIG. **3**, HO preparation is performed between the serving NG-RAN **21** and the target NG-RAN **22** in operation **300**, and Xn handover may be performed through HO execution in operation **301**. That is, the UE **10** may perform a handover procedure from the serving NG-RAN **21**, which is the base station that was initially provided with the service to the target NG-RAN **22**.

(54) The target NG-RAN **22** may transmit an N2 path switch request message to the AMF **101** in operation **302** in order to move a data path flowing to the existing serving NG-RAN **21** to itself. In this case, the N2 path switch request message may be transmitted to the AMF **101** by including the N2 SM message sending to each SMF **107** or MB-SMF **112** for each session. In this case, the target NG-RAN **22** may include an MBS capability indicating whether it supports 5 MBS in the N2 path switch request message.

(55) The SM message may include tunnel endpoint information (address information of the target NG-RAN, tunnel ID, and the like) for the target NG-RAN **22** to receive data traffic for the corresponding session.

(56) Upon receiving the request message in operation **302**, the AMF **101** may transmit an Nsmf_PDUSession_UpdateSMcontext request message to the corresponding SMF **107** in operation **303**. Here, the Nsmf_PDUSession_UpdateSMcontext request message may include a purpose for the AMF **101** to transmit the SM message to the SMF **107**. Further, the Nsmf_PDUSession_UpdateSMcontext request message may include a 5 MBS capability of the target NG-RAN **22** from capability information of the NG-RAN already known by the AMF **101** through a configuration procedure separately from the N2 SM message.

(57) The SMF **107** may determine to use a shared delivery method as a data traffic delivery method corresponding to the MBS session ID in operation **304** according to the 5 MBS capability of the target NG-RAN **22** delivered in operation **303**. The SMF **107** may know a 5 MBS capability through the capability transmitted by the target NG-RAN **22** with a method of identifying the 5 MBS capability of the target NG-RAN **22**. The SMF **107** may know a 5 MBS capability by distinguishing whether an assigned tunnel ID in the tunnel endpoint information sent by the target NG-RAN **22** is an ID for an individual tunnel or an ID for a shared tunnel with another method of identifying the 5 MBS capability of the target NG-RAN **22**. That is, an ID range for the shared tunnel and an ID range for the individual tunnel may be operated separately or an information

element (IE) for ID delivery for the shared tunnel may be distinguished by using an IE separately from the ID for the individual tunnel.
(58) Therefore, when a 5 MBS capability of the target NG-RAN **22** is not supported, it is determined to individual delivery, and when a 5 MBS capability of the target NG-RAN **22** is supported, it is determined to shared delivery. Accordingly, the SMF **107** may transmit an N4 session modification request message to the UPF **108** for all PDU sessions in operation **305**, and receive an N4 session modification response message from the UPF **108** in operation **306**. Further, the UPF **108** may generate an N3 end marker message to transmit the N3 end marker message to the serving NG-RAN **21** in operation **307**, thereby notifying the target NG-RAN **22** of the change of the path. Further, such an N3 end marker message is transmitted from the serving NG-RAN **21** to the target NG-RAN **22** in operation **307-1**, so that both the serving NG-RAN **21** and the target NG-RAN **22** may recognize that the path has been changed.
(59) For the PDU session connected to the MBS session ID, when the SMF **107** transmits an Nsmf_PDUSession_UpdateSMcontext response message to the AMF **101** in operation **308** in response to the Nsmf_PDUSession_UpdateSMcontext request message in operation **303**, the N2 SM message sending to the target NG-RAN **22** may include a shared delivery request, MB-SMF ID, temporary mobile group identity (TMGI) information, and MBS session ID. The information may be transmitted to the target RAN **22** through an N2 path switch request ACK message transmitting again by the AMF **101** in operation **309** in response to operation **302**. Accordingly, the target NG-RAN **22** may transmit a release resource message to the serving NG-RAN **21** in operation **310**.
(60) Upon receiving the N2 path switch request ACK message in operation **309**, the target NG-RAN **22** may transmit a message including the N2 SM message for generating a shared tunnel to the AMF **101** in operation **311** when there is no shared tunnel for the corresponding MBS session ID. The N2 SM message may include at least one of an MBS session ID, TMGI, MB-SMF ID, or target NG-RAN shared tunnel endpoint information (target NG-RAN address, shared tunnel ID, and the like). The MB-SMF ID is delivered to the AMF **101** together with the N2 SM message separately from the N2 SM message so that the AMF **101** may forward the N2 SM message to the MB-SMF **112** corresponding to the MB-SMF ID. Accordingly, in operation **312**, the AMF **101** may transmit the Nsmf_MBSSession_UpdateSMcontext request message including the N2 SM message to the MB-SMF **112** by including the corresponding MBS session ID. Accordingly, the MB-SMF **112** may switch MBS data transmitting to the target NG-RAN **22** to shared delivery. For such a shared tunnel, the MB-SMF **112** may generate a tunnel toward the target-RAN **22** for the corresponding MBS session ID through the N4 session modification process to the MB-UPF **111** in operation **313**. Further, the MB-SMF **112** may transmit responses to the requests in operations **312** and **311** to complete the handover procedure in operation **313**.
(61) According to another embodiment other than the method described above, when the target NG-RAN **22** has a context for the MBS session, and when the target NG-RAN **22** has a shared tunnel for the MBS session, there is no need to perform a separate handover procedure for the corresponding MBS session.
(62) However, when the target NG-RAN **22** receives (stores) a context for the MBS session from the serving NG-RAN **21**, if the target NG-RAN **22** does not have a shared tunnel for the MBS session, it may be switched to multicast through shared delivery to the target NG-RAN **22** according to the procedure of FIG. **3** described above.
(63) As another method, when the target NG-RAN **22** supports a 5 MBS capability in operation **302** of FIG. **3** described above, and when the target NG-RAN **22** receives (stores) a context for the MBS session from the serving NG-RAN **21** and knows an address or ID of the MB-SMF **112** serving the corresponding MBS session or TMGI, the target NG-RAN **22** transmits the N2 SM request message for the corresponding MBS session included in the N2 path switch request message of operation **302** to the AMF **101** (the same as operation **311**), the AMF **101** transmits the

N2 SM request message to the MB-SMF **112**, and the MB-SMF **112** transmits the N2 SM request message to the MB-UPF **111** to setup for shared delivery (the same as operations **312** and **313**). Therefore, because the shared tunnel was generated, the messages of operations **308** and **309** do not include a shared delivery request, MB-SMF ID, and TMGI information, and operations **311** to **313** are not performed. Xn handover is completed by performing procedures for operations **308** to **310** for a PDU session.

(64) Through the above method, when the target NG-RAN has an MBS capability, the target NG-RAN may smoothly provide the multicast service to the UE through shared delivery.

(65) FIG. **4** illustrates a process of supporting Xn handover to a UE in a multicast service according to an embodiment of the disclosure.

(66) In the following description, a serving NG-RAN, which is a serving base station, will be described with reference numeral **21**, and a target NG-RAN, which is the target for handover, will be described with reference numeral **22**. Further, in the disclosure, there may be a plurality of SMFs, and accordingly, they are indicated as SMF(s). However, it should be noted that reference numeral **107** illustrated in FIG. **1** is used for the SMF for convenience of description. Further, a plurality of UPFs may be used in the disclosure. It is noted that reference numeral **108** illustrated in FIG. **1** is used for the UPF for convenience of description.

(67) Referring to FIG. **4**, HO preparation may be performed between the serving NG-RAN **21** and the target NG-RAN **22** in operation **400**, and Xn handover may be performed through HO execution in operation **401**. That is, the UE **10** may perform a handover procedure from the serving NG-RAN **21**, which is the base station that was initially provided with the service to the target NG-RAN **22**.

(68) The target NG-RAN **22** may transmit an N2 path switch request message to the AMF **101** in operation **402** in order to move a data path flowing to the existing serving NG-RAN **21** to itself. In this case, the N2 path switch request message may be transmitted to the AMF **101** by including the N2 SM message sending to each SMF **107** or MB-SMF **112** for each session.

(69) The SM message may include tunnel endpoint information (address information of the target NG-RAN, tunnel ID, and the like) for the target NG-RAN **22** to receive data traffic for the corresponding session.

(70) The AMF **101** that has received the N2 path switch request message in operation **402** may transmit an Nsmf_PDUSession_UpdateSMcontext request message to the SMF **107** in order to deliver the SM message sending to the SMF **107** in operation **403** to the corresponding SMFs **107**.

(71) When the AMF **101** requests an MBS context so as to obtain information for selecting an MBS delivery mode in operation **403** or when the SMF **107** determines (identifies) that the AMF **101** needs to select an MBS delivery mode in operation **403**, the SMF **107** may transmit an Nsmf_PDUSession_UpdateSMcontext response message to the AMF **101** in operation **404**. The Nsmf_PDUSession_UpdateSMcontext response message may include an MBS context including an MBS session ID or TMGI and ID or address information of an MB-SMF for an MBS session associated with a PDU session.

(72) When determining the delivery mode for the connected PDU session corresponding to the MBS session delivered in operation **402**, the AMF **101** may recognize that the target NG-RAN **22** does not have a 5 MBS capability through information already configured with the target NG-RAN **22**. In this case, the AMF **101** may determine to support the service for the corresponding MBS session through individual delivery in operation **405**.

(73) In this case, while the AMF **101** transmits an Nsmf_PDUSession_UpdateSMcontext request message to the SMF **107** in operation **406** for the connected PDU session for individual delivery, the AMF **101** may transmit a change request to individual delivery for the MBS session ID or PDU session connected to the TMGI to the SMF **107**.

(74) Operations **411**, **412** and **413** are shown in **410** in FIG. **4**. Accordingly, in operation **411**, the SMF **107** may transmit the Nsmf_MBSSession_UpdateSMcontext request message to the MB-

SMF **112** by including the corresponding MBS session ID. Alternatively, the request in operation **406** may include tunnel endpoint information (e.g., UPF address and tunnel ID) of a PSA-UPF for receiving MBS traffic corresponding to the MBS session ID. In this case, the SMF **107** may include an associated PDU session ID connected with the MBS session ID.
(75) Upon receiving the Nsmf_MBSSession_UpdateSMcontext request message through operation **411**, the MB-SMF **112** may transmit the received tunnel endpoint information of the PSA-UPF to the MB-UPF **111** in order to deliver traffic corresponding to the MBS session ID to the PSA-UPF in operation **412**. The MB-UPF **111** may provide tunnel endpoint information (e.g., MB-UPF address, tunnel ID, lower layer MAC address, and the like) thereof to the MB-SMF **112** in operation **412** based on the received information. The tunnel endpoint information of the MB-UPF **111** may be provided to the SMF **107** by the MB-SMF **112** in response to the request in operation **406** described above in operation **413**.
(76) Upon receiving the tunnel endpoint information of the MB-UPF **111** through operation **413**, the SMF **107** may update the PDU session ID connected with the MBS session ID for individual delivery. To this end, in operation **420**, the SMF **107** may transmit an N4 session modification request message to the UPF **108**. The UPF **108** may generate an N4 session modification response message to transmit the N4 session modification response message to the SMF **107** in operation **421**. Operations **420** and **421** may be a case in which the UPF **108** operates as a PSA-UPF. Therefore, an individual tunnel between the PSA-UPF and the MB-UPF **111** may be generated for the PDU session ID to which the UE **10** is connected through operations **420** and **421**.
(77) Further, in order to generate an individual tunnel between the target NG-RAN **22** and the PSA-UPF, the SMF **107** may transmit the tunnel endpoint information of the target NG-RAN **22** received in operation **403** or **406** to the UPF **108** in operation **421**.
(78) N4 session modification, which transmits the tunnel endpoint information of the target NG-RAN **22** received by the SMF **107** to the UPF **108** in operation **403** or **406** to set up an individual tunnel with the target NG-RAN **22** may be performed separately after operation **404**.
(79) In operations **422** and **422-1**, the UPF **108** may transmit an N3 end marker to the target NG-RAN **22** through the serving NG-RAN **21**. That is, the UPF **108** completes performing the remaining Xn handover, so that when the target NG-RAN **22** does not have an MBS capability, the target NG-RAN **22** may smoothly provide a multicast service to the UE **10** through individual delivery.
(80) Thereafter, in operation **423**, the SMF **107** may generate and transmit an Nsmf_PDUSession_UpdateSMcontext response message in response to the Nsmf_PDUSession_UpdateSMcontext request message in operation **406**. Therefore, the AMF **101** may transmit an N2 path switch request ACK message to the target NG-RAN **22** in operation **424** to notify that the remaining Xn handover has been completed.
(81) Accordingly, the target NG-RAN **22** may transmit a release resource message to the source NG-RAN **21** to instruct the source NG-RAN **21** to release the resource allocated to the UE **10** in operation **425**.
(82) FIG. **5** illustrates a process of supporting Xn handover to a UE in a multicast service according to an embodiment of the disclosure.
(83) Referring to FIG. **5**, as previously described in FIG. **2**, a serving NG-RAN, which is a serving base station, will be described with reference numeral **21**, and a target NG-RAN, which is the target for handover, will be described with reference numeral **22**. Further, in the disclosure, there may be a plurality of SMFs, and accordingly, they are indicated as SMF(s). However, it should be noted that reference numeral **107** illustrated in FIG. **1** is used for the SMF for convenience of description. Further, a plurality of UPFs may be used in the disclosure. It is noted that reference numeral **108** illustrated in FIG. **1** is used for the UPF for convenience of description.
(84) With reference to FIG. **5**, HO preparation may be performed between the serving NG-RAN **21** and the target NG-RAN **22** in operation **500**. Further, in operation **501**, Xn handover may be

performed through HO execution. That is, the UE **10** may perform a handover procedure from the serving NG-RAN **21**, which is the base station that was initially provided with the service to the target NG-RAN **22**.
(85) The target NG-RAN **22** may transmit an N2 path switch request message to the AMF **101** in operation **502** in order to deliver a data path flowing to the existing serving NG-RAN **21** to itself. In this case, the N2 path switch request message may be transmitted to the AMF **101** by including the N2 SM message sending to each SMF **107** or MB-SMF **112** for each session.
(86) The SM message may include tunnel endpoint information (address information of the target NG-RAN, tunnel ID, and the like) for the target NG-RAN **22** to receive data traffic for the corresponding session.
(87) The AMF **101** that has received the request in operation **502** may transmit an Nsmf_PDUSession_UpdateSMcontext request message to the corresponding SMFs **107** in order to deliver the SM message sending to the SMF **107** in operation **503**.
(88) When the AMF **101** requests an MBS context so as to obtain information for selecting an MBS delivery mode in operation **503** or when the SMF **107** recognizes (determines or identifies) that the MBS delivery mode selection is necessary to the AMF **101** in operation **503**, the SMF **107** may transmit an Nsmf_PDUSession_UpdateSMcontext response message to the AMF **101** in operation **504**. The Nsmf_PDUSession_UpdateSMcontext response message may include an MBS context including an MBS session ID or TMGI and ID or address information of MB-SMF for an MBS session associated with a PDU session.
(89) When determining a delivery mode for the connected PDU session corresponding to the MBS session received through operation **504**, the AMF **101** may recognize that the target NG-RAN **22** has a 5 MBS capability through information already configured with the target NG-RAN **22**. In this case, the AMF **101** may determine to support the service for the corresponding MBS session through shared delivery in operation **505**.
(90) In operation **506**, the AMF **101** may include information on shared delivery in the Nsmf_PDUSession_UpdateSMcontext request message and transmit the Nsmf_PDUSession_UpdateSMcontext request message to the SMF **107**. Therefore, the SMF **107** may include information on shared delivery in the N4 session modification request message and transmitting the message to the UPF **108** in operation **507**, thereby notifying the UPF **108** of this. Upon receiving the N4 session modification request message in operation **507**, the UPF **108** may generate an N4 session modification response message to respond to the SMF **107** in operation **508**. Further, the UPF **108** may transmit an N3 End marker to the target NG-RAN **22** in operations **509** and **509-1** through the source NG-RAN **21** based on the N4 session modification request message received in operation **507**.
(91) The SMF **107** may generate an Nsmf_PDUSession_UpdateSMcontext response message in response to the Nsmf_PDUSession_UpdateSMcontext request message in operation **506** and transmit the Nsmf_PDUSession_UpdateSMcontext response message to the AMF **101** in operation **510**.
(92) After receiving the Nsmf_PDUSession_UpdateSMcontext response message in operation **510**, the AMF transmits an N2 path switch request ACK message to the target NG-RAN **22** in operation **511**, thereby transmitting a shared delivery request, information on an MBS session ID or TMGI, and MB-SMF ID to the target RAN.
(93) The target NG-RAN **22** that has received the N2 path switch request ACK message transmits a release resource message to the source NG-RAN **21** in operation **512**, and enables the source NG-RAN **21** to release a resource allocated to the UE **10**. Thereafter, when there is no shared tunnel for the corresponding MBS session ID based on the message received in operation **511**, in order to generate a shared tunnel, the target NG-RAN **22** may transmit a message including the N2 SM message to the AMF **101** in operation **513**. The N2 SM message may include an MBS session ID or TMGI, MB-SMF ID, and shared tunnel endpoint information of the target NG-RAN (address of the

target NG-RAN, shared tunnel ID, and the like). Further, the MB-SMF ID is delivered to the AMF **101** together with the N2 SM message separately from the N2 SM message, so that the AMF **101** may forward the N2 SM message to the MB-SMF **112** corresponding to the MB-SMF ID in operation **514**. Accordingly, in operation **514**, the AMF **101** may transmit an Nsmf_MBSSession_UpdateSMcontext request message including the N2 SM message to the MB-SMF **112** by including the corresponding MBS session ID. Accordingly, the MB-SMF **112** switches data transmission corresponding to the MBS session ID to shared delivery, and in order to generate a shared tunnel, the MB-SMF **112** may generate a tunnel toward the target-RAN **22** for the corresponding MBS session ID through a process of transmitting an N4 session modification request to the MB-UPF **111** and receiving an N4 session modification response in operations **515** and **515-1**. Further, the MB-SMF **112** may generate a response to operation **514** and transmit the response to the AMF **101**, and the AMF **101** may generate a response to the request in operation **513** to transmit the response to the target NG-RAN **22**, thereby completing a handover procedure.
(94) According to a modified embodiment of the form described in FIG. **5**, when the target NG-RAN **22** has a warning having a context for the MBS session and a shared tunnel for the MBS session in the target NG-RAN **22**, there is no need to perform a separate handover procedure for the corresponding MBS session.
(95) However, when the target NG-RAN **22** receives (stores) a context for the MBS session from the serving NG-RAN **21**, and there is no shared tunnel for the MBS session in the target NG-RAN **22**, it may be switched to multicast through shared delivery to the target NG-RAN **22** according to the procedure of FIG. **5**.
(96) As another example, in operation **502** of FIG. **5**, because the target NG-RAN **22** supports a 5 MBS capability and receives (stores) a context for the MBS session from the serving NG-RAN **21**, when the target NG-RAN **22** knows an address or ID of MB-SMF serving the corresponding MBS session or TMGI, the target NG-RAN **22** transmits an N2 SM request message for the corresponding MBS session and the N2 path switch request message in operation **502** to the AMF **101** (the same as operation **513**), the AMF **101** transmits the N2 SM request message to the MB-SMF **112**, and the MB-SMF **112** transmits the N2 SM request message to the MB-UPF **111** to perform setup for shared delivery (the same as operations **514** and **515**). Therefore, because the shared tunnel was generated, the message in operation **511** does not include a shared delivery request, MB-SMF ID, TMGI information, and the like, and operations **513** to **515** are not performed. Xn handover may be completed by performing the procedure for operations **510** to **512** for a PDU session.
(97) Through the above method, when the target NG-RAN **22** has an MBS capability, the target NG-RAN **22** may smoothly provide the multicast service to the UE **10** through shared delivery.
(98) FIG. **6** illustrates a process of supporting N2 handover to a UE in a multicast service according to an embodiment of the disclosure.
(99) In the following description, a serving NG-RAN, which is a serving base station, will be described with reference numeral **21**, and a target NG-RAN, which is the target for handover, will be described with reference numeral **22**. Further, in the disclosure, there may be a plurality of SMFs, and accordingly, they are indicated as SMF(s). However, it should be noted that reference numeral **107** illustrated in FIG. **1** is used for the SMF for convenience of description. Further, a plurality of UPFs may be used in the disclosure. It is noted that reference numeral **108** illustrated in FIG. **1** is used for the UPF for convenience of description. Further, in the flowchart of FIG. **6**, the serving AMF and the target AMF to be handover exist. Accordingly, reference numeral **101***a* is assigned to the serving AMF, and reference numeral **101***b* is assigned to the target AMF.
(100) Referring to FIG. **6**, when the UE **10** receiving a multicast service is determined to perform N2-based handover in the serving NG-RAN **21** during a handover process, the UE **10** prepares handover through a process up to operation **632** illustrated in FIG. **6** through a handover preparation process, and when the preparation is completed, handover may be performed by

actually executing the handover through HO execution and through operation **640**. This will be described in detail through the following description.
(101) The serving NG-RAN **21** may transmit an HO required message to the serving AMF **101***a* in operation **601**. Accordingly, the serving AMF **101***a* may select the target AMF **101***b* to be moved to in operation **602**. The serving AMF **101***a* may transmit a Namf_communication_CreatUEContext request message to the target AMF **101***b* in operation **603**. Thereafter, the target AMF **101***b* that has received the Namf_communication_CreatUEContext request message may transmit an Nsmf_PDUSession_UpdateSMcontext request message to the corresponding SMF **107** for each PDU session in operation **604**. Therefore, the SMF **107** may perform UPF re-allocation, if required in operation **605**. The SMF **107** that has performed operation **605** or that has not performed operation **605** because it is not necessary, may transmit a Namf_communication_CreatUEContext response message to the target AMF **101***b* in operation **606**. Accordingly, the target AMF **101***b* may recognize that handover of the UE **10** receiving the MBS has been requested.
(102) The target AMF **101***b* may transmit a handover request to the target NG-RAN **22** in operation **607**. The target NG-RAN **22** may generate a response to the handover request to transmit the response to the target AMF **101***b* in operation **608**. When the target NG-RAN **22** responds with a Handover request ACK, the target NG-RAN **22** may notify the target AMF **101***b* of whether the target NG-RAN **22** supports a 5 MBS capability. The target NG-RAN **22** may transmit an N2 SM message sending to each SMF **107** for each PDU session to the target AMF **101***b*, in this case, the N2 SM message may include an MBS capability indicating whether the target NG-RAN **22** supports 5 MBS.
(103) The SM message may include tunnel endpoint information (address information of the target NG-RAN, tunnel ID, and the like) for the target NG-RAN **22** to receive data traffic for the corresponding session.
(104) The target AMF **101***b* that has received the request in operation **608** may transmit an Nsmf_PDUSession_UpdateSMcontext request message to the corresponding SMFs **107** in order to deliver the SM message sending to the SMF **107** in operation **609**. The Nsmf_PDUSession_UpdateSMcontext request message may include a 5 MBS capability of the target NG-RAN **22** from capability information of the NG-RAN already known by the target AMF **101***b* through a configuration operation separately from the N2 SM message.
(105) The SMF **107** may determine to use an individual delivery method as a data traffic delivery method corresponding to the MBS session ID in operation **610** according to the 5 MBS capability of the target NG-RAN **22**. The SMF **107** may know a 5 MBS capability through the capability sent by the target NG-RAN **22** with a method of identifying the 5 MBS capability of the target NG-RAN **22** or may know a 5 MBS capability by distinguishing whether an assigned tunnel ID in information of the tunnel endpoint sending by the target NG-RAN **22** is an ID for an individual tunnel or an ID for a shared tunnel. That is, an ID range for the shared tunnel and an ID range for the individual tunnel may be operated separately or an information element (IE) for ID delivery for the shared tunnel may be distinguished by using an IE separately from the ID for the individual tunnel.
(106) Therefore, when the 5 MBS capability of the target NG-RAN **22** is not supported, it may be determined to individual delivery, and when the 5 MBS capability of the target NG-RAN **22** is supported, it may be determined to shared delivery.
(107) Operations **621**, **622**, **623**, **624** and **625** are shown in **620** in FIG. **6**. Accordingly, in operation **621**, the SMF **107** may transmit an Nsmf_MBSSession_UpdateSMcontext request message to the MB-SMF **112** by including the corresponding MBS session ID. Alternatively, the Nsmf_MBSSession_UpdateSMcontext request message may include tunnel endpoint information (e.g., UPF address and tunnel ID) of the PSA-UPF for receiving MBS traffic corresponding to the MBS session ID. In this case, the SMF **107** may include an associated PDU session ID connected with the MBS session ID.

(108) Upon receiving the information, the MB-SMF **112** transmits the received tunnel endpoint information of the PSA-UPF to the MB-UPF **111** so that traffic corresponding to the MBS session ID may be transmitted to the PSA-UPF in operation **622**, and the MB-UPF **111** may transmit tunnel endpoint information (e.g., MB-UPF address, tunnel ID, lower layer MAC address, and the like) thereof to the MB-SMF **112**. The information may be transmitted to the SMF **107** in response to the request in operation **621** in operation **623**. Accordingly, the SMF **107** that has received tunnel endpoint information of the MB-UPF **111** may update a PDU session ID connected with the MBS session ID for individual delivery and for update, in operation **624**, the SMF **107** may transmit the PDU session ID to the PSA-UPF to generate an individual tunnel between the PSA-UPF and the MB-UPF for the connected PDU session ID.
(109) In order to generate an individual tunnel between the target NG-RAN **22** and the PSA-UPF, the SMF **107** may transmit the tunnel endpoint information of the target NG-RAN **22** received in operation **609** to the UPF **108** in operation **624**.
(110) N4 session modification request in operation **624**, which delivers the tunnel endpoint information of the target NG-RAN **22** received in operation **609** to the UPF **108** to set an individual tunnel with the target NG-RAN **22** and receive the N4 session modification response in operation **625**, may be separately performed in operation **640**.
(111) When the remaining N2 handover preparation process is completed in operations **631** and **632** and the HO execution process is performed in operation **640**, if the target NG-RAN **22** does not have an MBS capability, the target NG-RAN **22** may smoothly provide a multicast service to the UE **10** through individual delivery.
(112) FIG. **7** illustrates a process of supporting N2 handover to a UE in a multicast service according to an embodiment of the disclosure.
(113) In the following description, a serving NG-RAN, which is a serving base station, will be described with reference numeral **21**, and a target NG-RAN, which is the target for handover, will be described with reference numeral **22**. Further, in the disclosure, there may be a plurality of SMFs, and accordingly, they are indicated as SMF(s). However, it should be noted that reference numeral **107** illustrated in FIG. **1** is used for the SMF for convenience of description. Further, a plurality of UPFs may be used in the disclosure. It is noted that reference numeral **108** illustrated in FIG. **1** is used for the UPF for convenience of description. Further, in the flowchart of FIG. **7**, the serving AMF and the target AMF to be handover exist. Accordingly, reference numeral **101***a* is assigned to the serving AMF, and reference numeral **101***b* is assigned to the target AMF.
(114) Referring to FIG. **7**, when the UE **10** receiving a multicast service is determined to perform N2-based handover in the serving NG-RAN **21** during a handover process, the UE **10** prepares handover through a handover preparation process and through the process up to operation **718**, and when the handover preparation is completed, handover may be performed through a process of actually executing handover through HO execution in operation **719**. This will be described in detail through the following description.
(115) The serving NG-RAN **21** may transmit an HO required message to the serving AMF **101***a* in operation **701**, and the serving AMF **101***a* may select a target AMF **101***b* to be moved in operation **702**. The serving AMF **101***a* may transmit a Namf_communication_CreatUEContext request message to the target AMF **101***b* in operation **703**, and perform operations **704**, **705** and **706** for the SMF **107** and the UPF **108** for each PDU session, and then the target AMF **101***b* may transmit a handover request message to the target NG-RAN **22** in operation **707**.
(116) In response to the request in operation **707**, the target NG-RAN **22** may generate a response message and transmit the response message to the target AMF **101***b* in operation **708**. When the target NG-RAN **22** responds with a handover request ACK, the target NG-RAN **22** may notify the target AMF **101***b* of whether the target NG-RAN **22** supports a 5 MBS capability. The target NG-RAN **22** may transmit the N2 SM message sending to each SMF **107** to the target AMF **101***b* for each PDU session. In this case, the target NG-RAN **22** may include whether an MBS capability is

supported indicating whether the target NG-RAN **22** supports 5 MBS.
(117) The SM message may include tunnel endpoint information (address information of the target NG-RAN, tunnel ID, and the like) for the target NG-RAN **22** to receive data traffic for the corresponding session.
(118) The target AMF **101***b* that has received the request may transmit an Nsmf_PDUSession_UpdateSMcontext request message to the corresponding SMFs **107** in order to deliver the SM message sending to the SMF **107** in operation **709**. The Nsmf_PDUSession_UpdateSMcontext request message may include a 5 MBS capability of the target NG-RAN **22** from capability information of the NG-RAN already known by the target AMF **101***b* through a configuration or the like separately from the N2 SM message.
(119) The SMF **107** may determine to use a shared delivery method as a data traffic delivery method corresponding to the MBS session ID in operation **710** according to the received 5 MBS capability of the target NG-RAN **22**. The SMF **107** may know a 5 MBS capability through the capability sending by the target NG-RAN **22** with a method of identifying the 5 MBS capability of the target NG-RAN **22** or may know a 5 MBS capability by distinguishing whether an assigned tunnel ID is an ID for an individual tunnel or an ID for a shared tunnel in information of the tunnel endpoint sending by the target NG-RAN **22**. That is, an ID range for the shared tunnel and an ID range for the individual tunnel may be operated separately or an information element (IE) for ID delivery for the shared tunnel may be distinguished by using an IE separately from the ID for the individual tunnel.
(120) Therefore, when the 5 MBS capability of the target NG-RAN **22** is not supported, it may be determined to individual delivery, and when the 5 MBS capability of the target NG-RAN **22** is supported, it may be determined to shared delivery.
(121) For the PDU session connected to the MBS Session ID, when the SMF **107** transmits an Nsmf_PDUSession_UpdateSMcontext response message in response to the Nsmf_PDUSession_UpdateSMcontext request message of operation **709** in operation **713**, the Nsmf_PDUSession_UpdateSMcontext response message may include at least one of a shared delivery request, MB-SMF ID, TMGI information, or MBS session ID as an N2 SM message sending to the target NG-RAN **22**. The information may be transmitted again to the target RAN **22** through the N2 SM request message of operation **714** by the target AMF **101***b*.
(122) When there is no shared tunnel for the corresponding MBS session ID, the target NG-RAN **22** that has received the information may transmit a message including an N2 SM message for generating a shared tunnel to the target AMF **101***b* in operation **715**. The N2 SM message may include at least one of the MBS session ID, TMGI, MB-SMF ID, or target NG-RAN shared tunnel endpoint information (target NG-RAN address, shared tunnel ID, and the like). Further, the MB-SMF ID is delivered to the target AMF **101***b* together with the N2 SM message separately from the N2 SM message, so that the target AMF **101***b* may forward the N2 SM message to the MB-SMF **112** corresponding to the MB-SMF ID. Accordingly, in operation **716**, the target AMF **101***b* may transmit an Nsmf_MBSSession_UpdateSMcontext request message including the N2 SM message to the MB-SMF **112** by including the corresponding MBS session ID. Accordingly, the MB-SMF **112** switches to the shared delivery, and for the shared tunnel for this, in operation **717**, the MB-SMF **112** may generate a tunnel toward the target-RAN **22** for the corresponding MBS session ID through an N4 session modification process to the MB-UPF **111**. Further, the MB-SMF **112** and the target AMF **101***b* may transmit respective responses to the requests in operations **716** and **715**. Because the target AMF **101***b* transmits a response to the message in operation **703** to the serving AMF **101***a* in operation **718**, the target AMF **101***b* may complete the handover preparation procedure and perform the HO execution procedure in operation **719**.
(123) According to another embodiment of FIG. **7**, when the target NG-RAN **22** has (stores) a context for the MBS session, and the target NG-RAN **22** has a shared tunnel for the MBS session, there is no need to perform a separate handover procedure for the corresponding MBS session.

(124) However, when the target NG-RAN **22** receives (stores) a context for the MBS session from the serving NG-RAN **21**, and there is no shared tunnel for the MBS session in the target NG-RAN **22**, it is possible to switch to multicast through shared delivery to the target NG-RAN according to the procedure of FIG. **7**.
(125) As another method, in operation **708** of FIG. **7**, because the target NG-RAN **22** supports a MBS capability and receives (stores) a context for the MBS session from the serving NG-RAN **21**, when the target NG-RAN **22** knows an address or ID of MB-SMF serving the MBS session or TMGI, the target NG-RAN **22** may transmit an N2 SM request message for the corresponding MBS session and the handover request ACK message in operation **708** to the target AMF **101***b* (the same as operation **715**), the target AMF **101***b* may transmit the N2 SM request message to the MB-SMF **112**, and the MB-SMF **112** may transmit the N2 SM request message to the MB-UPF **111** to perform setup for shared delivery (the same as operations **716** and **717**). Therefore, because the shared tunnel was generated, the MB-SMF ID and TMGI information are not included in the shared delivery request in the messages of operations **713** and **714**, and operations **715** to **717** are not performed. HO preparation for N2 handover may be completed by performing the procedures for operations **711**, **712**, **713** and **718** for a PDU session.
(126) After the HO preparation is completed as described above, in operation **719**, the serving AMF **101***a* transmits a HO command to the serving NG-RAN **21**, thereby executing the HO execution process.
(127) Through the above method, when the target NG-RAN **22** has an MBS capability, the target NG-RAN may smoothly provide the multicast service to the UE through shared delivery.
(128) FIG. **8** illustrates a process of supporting N2 handover to a UE in a multicast service according to an embodiment of the disclosure.
(129) In the following description, a serving NG-RAN, which is a serving base station, will be described with reference numeral **21**, and a target NG-RAN, which is the target for handover, will be described with reference numeral **22**. Further, in the disclosure, there may be a plurality of SMFs, and accordingly, they are indicated as SMF(s). However, it should be noted that reference numeral **107** illustrated in FIG. **1** is used for the SMF for convenience of description. Further, a plurality of UPFs may be used in the disclosure. It is noted that reference numeral **108** illustrated in FIG. **1** is used for the UPF for convenience of description. Further, in the flowchart of FIG. **8**, the serving AMF and the target AMF to be handover exist. Accordingly, reference numeral **101***a* is assigned to the serving AMF, and reference numeral **101***b* is assigned to the target AMF.
(130) Referring to FIG. **8**, when the UE **10** receiving a multicast service is determined to perform N2-based handover in the serving NG-RAN **21** during a handover process, the UE **10** prepares handover through a handover preparation process and through a process up to operation **827**, and when handover preparation is completed, handover occurs through a process of actually executing handover through HO execution in operation **828**. Hereinafter, this will be described in detail through the following description.
(131) The serving NG-RAN **21** may transmit an HO required message to the serving AMF **101***a* in operation **801**, and the serving AMF **101***a* may select the target AMF **101***b* to be moved in operation **802**. The serving AMF **101***a* transmits the Namf_communication_CreatUEContext request message to the target AMF **101***b* in operation **803**, and in order to deliver the SM message sending to the SMF **107** in operation **804** to the SMFs **107** managing each PDU session, the target AMF **101***b* may transmit to the SMF **107** the Nsmf_PDUSession_UpdateSMcontext request message. The SMF **107** may perform UPF re-allocation, if required in step operation **805**.
(132) When the target AMF **101***b* requests an MBS context in order to obtain information for selecting an MBS delivery mode in operation **804** or when the SMF **107** recognizes that the target AMF **101***b* needs to select an MBS delivery mode in operation **804**, the SMF **107** may transmit an Nsmf_PDUSession_UpdateSMcontext response to the target AMF **101***b* in operation **806**. In this case, the Nsmf_PDUSession_UpdateSMcontext response message may include an MBS context

including an MBS session ID or TMGI and an ID of the MB-SMF **112** or address information of the MB-SMF **112** for an MBS session associated with a PDU session.
(133) Through operations **807** and **808**, the target AMF **101***b* may transmit a handover request message to the target NG-RAN **22** and receive a response thereof. In determining a delivery mode for the MBS session connected to the PDU session, the target AMF **101***b* may recognize that the target NG-RAN **22** does not have a 5 MBS capability through information already configured with the target NG-RAN **22** and in operation **809**, the target AMF **101***b* may determine to support the service for the corresponding MBS session through individual delivery.
(134) In this case, the target AMF **101***b* may transmit a change request message to individual delivery for a PDU session connected to the TMGI or the MBS session ID to the SMF **107** while transmitting the Nsmf_PDUSession_UpdateSMcontext request message to the SMF **107** in operation **810** for the connected PDU session for individual delivery.
(135) Operations **821**, **822**, **823**, **824** and **825** are shown in **820** in FIG. **8**. Accordingly, in operation **821**, the SMF **107** may transmit an Nsmf_MBSSession_UpdateSMcontext request message to the MB-SMF **112** by including the corresponding MBS session ID. Further, the Nsmf_MBSSession_UpdateSMcontext request message may include tunnel endpoint information (e.g., UPF address and tunnel ID) of a PSA-UPF for receiving MBS traffic corresponding to the MBS session ID. In this case, the SMF **107** may include an associated PDU session ID connected with the MBS session ID.
(136) Upon receiving the information, the MB-SMF **112** may transmit the received tunnel endpoint information of the PSA-UPF to the MB-UPF **111** in order to transmit traffic corresponding to the MBS session ID to the PSA-UPF and transmit tunnel endpoint information (e.g., MB-UPF address, tunnel ID, lower layer MAC address, and the like) of the MB-UPF **111** to the MB-SMF **112** in operation **822**. In operation **823**, the MB-SMF **112** may transmit the information to the SMF **107** in response to the request in operation **821**. Accordingly, the SMF **107** that has received the tunnel endpoint information of the MB-UPF **111** may update the PDU session ID connected with the MBS session ID for individual delivery, and for this purpose, in operations **814** and **815**, the SMF **107** may transmit the PDU session ID to the PSA-UPF to generate an individual tunnel between the PSA-UPF and the MB-UPF for the connected PDU session ID.
(137) In order to generate an individual tunnel between the target NG-RAN **22** and the PSA-UPF, the SMF **107** may transmit the tunnel endpoint information of the target NG-RAN **22** received in operations **808** and **810** to the UPF **108** in operation **824** and receive the N4 session modification response in operation **825**.
(138) The UPF **108** transmits the tunnel endpoint information of the target NG-RAN **22** received in operations **808** and **810** to the UPF **108**; thus, N4 session modification for setting an individual tunnel with the target NG-RAN **22** may be performed separately.
(139) In operations **826** and **827**, the SMF **107** and the target AMF **101***b* may complete the remaining N2 handover preparation process and execute an HO execution process in operation **828**, so that when the target NG-RAN **22** does not have an MBS capability, the target NG-RAN **22** may smoothly provide a multicast service to the UE **10** through individual delivery.
(140) FIG. **9** illustrates a process of supporting N2 handover to a UE in a multicast service according to an embodiment of the disclosure.
(141) In the following description, a serving NG-RAN, which is a serving base station, will be described with reference numeral **21**, and a target NG-RAN, which is the target for handover, will be described with reference numeral **22**. Further, in the disclosure, there may be a plurality of SMFs, and accordingly, they are indicated as SMF(s). However, it should be noted that reference numeral **107** illustrated in FIG. **1** is used for the SMF for convenience of description. Further, a plurality of UPFs may be used in the disclosure. It is noted that reference numeral **108** described with reference to FIG. **1** is used for the UPF for convenience of description. Further, in the flowchart of FIG. **9**, the serving AMF and the target AMF to be handover exist. Accordingly,

reference numeral **101***a* is assigned to the serving AMF, and reference numeral **101***b* is assigned to the target AMF.

(142) Referring to FIG. **9**, when the UE **10** receiving a multicast service is determined to perform N2-based handover in the serving NG-RAN **21** during a handover process, the UE **10** prepares handover through a process up to operation **917** and through a handover preparation process, when handover preparation is completed, handover occurs through a process of actually executing handover through HO execution in operation **918**. This will be described in detail through the following description.

(143) The serving NG-RAN **21** may transmit an HO required message to the serving AMF **101***a* in operation **901**, and the serving AMF **101***a* may select a target AMF **101***b* to be moved in operation **902**. In operation **903**, the serving AMF **101***a* transmits the Namf_communication_CreatUEContext request message to the target AMF **101***b*, and the target AMF **101***b* may transmit an Nsmf_PDUSession_UpdateSMcontext request message to the SMFs **107** managing each PDU session to the SMF **107** in order to deliver the SM message to the SMF **107** in operation **904**. The SMF **107** may perform UPF re-allocation, if required in step operation **905**.

(144) When the target AMF **101***b* requests an MBS context in order to obtain information for selecting an MBS delivery mode in operation **904** or when the SMF **107** recognizes that it is necessary to select an MBS delivery mode to the target AMF **101***b* in operation **904**, the SMF **107** may transmit an Nsmf_PDUSession_UpdateSMcontext response message to the target AMF **101***b* in operation **906**. The Nsmf_PDUSession_UpdateSMcontext response message may include an MBS context including at least one of an MBS session ID, TMGI, an ID of the MB-SMF **112**, or address information of the MB-SMF for an MBS session associated with the PDU session.

(145) Through operations **907** and **908**, the target AMF **101***b* may request handover to the target NG-RAN **22** and receive a response thereof from the target NG-RAN **22**. In determining a delivery mode for the MBS session connected to the PDU session, the target AMF **101***b* may recognize that the target NG-RAN **22** supports a 5 MBS capability through information already configured with the target NG-RAN **22**, and in operation **909**, the target AMF **101***b* may determine to support a service for the corresponding MBS session through shared delivery.

(146) In this case, the target AMF **101***b* may transmit at least one of information of a shared delivery request, MBS Session ID, TMGI, or MB-SMF ID for the MBS session to the target RAN **22** through the N2 SM request message in operation **914**.

(147) When there is no shared tunnel for the corresponding MBS session ID, the target NG-RAN **22** that has received the information may transmit a message including an N2 SM message for generating a shared tunnel to the target AMF **101***b* in operation **915**. The N2 SM message may include at least one of an MBS session ID, TMGI, MB-SMF ID, or target NG-RAN shared tunnel endpoint information (target NG-RAN address, shared tunnel ID, and the like), and the MB-SMF ID is delivered to the target AMF **101***b* together with the N2 SM message separately from the N2 SM message, so that the target AMF **101***b* may forward the N2 SM message to the MB-SMF **112** corresponding to the MB-SMF ID. Accordingly, in operation **916**, the target AMF **101***b* may transmit an Nsmf_MBSSession_UpdateSMcontext request message including the N2 SM message to the MB-SMF **112** by including the corresponding MBS session ID. Accordingly, the target AMF **101***b* may switch to shared delivery, and for a shared tunnel for shared delivery, the MB-SMF **112** may generate a tunnel toward the target-RAN **22** for the corresponding MBS session ID through an N4 session modification process to the MB-UPF **111** in operation **917**. Further, the MB-SMF **112** and the target AMF **101***b* may transmit respective responses to operations **916** and **915**, in operation **918**, and because the target AMF **101***b* transmits a response to the message in operation **903** to the serving AMF **101***a* in operations **917** and **918**, the target AMF **101***b* may complete a handover preparation procedure and perform an HO execution procedure of operation **919**.

(148) According to another embodiment of the method illustrated in FIG. **9**, when the target NG-

RAN **22** has a context for the MBS session, if the target NG-RAN **22** has a shared tunnel for the MBS session, it is not necessary to perform a separate handover procedure for the corresponding MBS session.
(149) However, the target NG-RAN **22** may receive (store) a context for the MBS session from the serving NG-RAN **21**, and when the target NG-RAN **22** does not have a shared tunnel for the MBS session, it may be switched to multicast through shared delivery to the target NG-RAN **22** according to the procedure of FIG. **9**.
(150) As another example, in operation **908** of FIG. **9**, because the target NG-RAN **22** supports a 5 MBS capability and receives (stores) a context for the MBS session from the serving NG-RAN **21**, when the target NG-RAN **22** knows an address or ID of MB-SMF serving TMGI and the corresponding MBS session, the target NG-RAN **22** may transmit an N2 SM request message for the corresponding MBS session and the handover request ACK message in operation **908** to the target AMF **101***b* (the same as operation **915**), the target AMF **101***b* may transmit the N2 SM request message to the corresponding MB-SMF **112**, and the MB-SMF **112** may transmit the N2 SM request message to the MB-UPF **111** to perform setup for shared delivery (the same as operations **916** and **917**). Therefore, because the shared tunnel was generated, the messages of operations **913** and **914** do not include MB-SMF ID and TMGI information such as a shared delivery request, and operations **915** to **917** are not performed. Meanwhile, HO preparation for N2 handover may be completed by performing the procedures for operations **910**, **911**, **912**, **913** and **918** for a PDU session.
(151) After the HO preparation is completed, as described above, in operation **919**, the serving AMF **101***a* transmits a HO command to the serving NG-RAN **21** to perform an HO execution process.
(152) Through the above method, when the target NG-RAN **22** has an MBS capability, the target NG-RAN **22** may smoothly provide a multicast service to the UE **10** through shared delivery.
(153) FIG. **10** is a block diagram illustrating a configuration of a UE to which the disclosure may be applied.
(154) With reference to FIG. **10**, the UE **10** may include a communication unit **1001**, a UE processor **1002**, and a UE memory **1003**. The communication unit **1001** may include a wireless communication module capable of communicating with a cellular system such as long term evolution (LTE), LTE-A, and 5G networks, a modem, and/or a communication processor.
(155) The UE processor **1002** may perform overall control of the UE **10**, and control to connect and release a call according to a user's request, and to provide a user's customized service. Further, the UE processor **1002** may receive MBS data traffic according to the disclosure, and may control directly an operation required for handover or may control using some configurations of the communication unit **1001**.
(156) The UE memory **1003** may include an area for storing various control information necessary for the UE **10** and an area for storing user data.
(157) Further, the UE **10** may have a display device such as a display, LCD and/or LED for an interface with a user, and include various interfaces for detecting a user input. Further, the UE **10** may be implemented to have more components, as needed.
(158) FIG. **11** is a block diagram illustrating a configuration of a network function (NF) device to which the disclosure may be applied.
(159) Referring to FIG. **11**, the NF device may include a network interface **1101**, a processor **1102**, and a memory **1103**. The network interface **1101** may provide an interface for communication with other NFs. For example, when the NF is an AMF, the network interface **1101** may provide an interface for communicating with the SMF. As another example, when the NF is an UPF, the network interface **1101** may provide an interface for transmitting and receiving various data/signals/messages to and from an RNA and/or AMP and/or MB-SMF.
(160) The processor **1102** may control an operation of the corresponding NF. For example, when

the NF is an AMF, the processor **1102** may control operations in FIGS. **2** to **9** described above. As another example, when the NF is the SMF, the processor **1102** may control an operation of the SMF during the operation according to FIGS. **2** to **9** described above. Further, when the NF is an MB-SMF/MB-AMF, the processor **1102** may control operations according to the above-described drawings in the same manner.
(161) The memory **1103** may store information for the control of the NF, information generated during the control, and information necessary according to the disclosure.
(162) According to the disclosure, even when a terminal using a multicast service moves in a 5G system (5GS), it is possible to use smoothly the multicast service.
(163) While the disclosure has been shown and described with reference to various embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure as defined by the appended claims and their equivalents.

Claims

1. A method for Xn handover of a multicast/broadcast service (MBS) by a session management function (SMF) in a mobile communication system, the method comprising: receiving, from a target next generation radio access network (NG-RAN) node through an access and mobility management function (AMF), a first message including information on whether the target NG-RAN node supports the MBS; determining an individual delivery method for MBS data, in case that the target NG-RAN node does not support the MBS based on the information received from the target NG-RAN node; and performing an establishment of individual MBS data delivery, wherein the information on whether the target NG-RAN node supports the MBS is N2 session management (SM) information received from the target NG-RAN node to the SMF via the AMF.
2. The method of claim 1, wherein the performing the establishment of the individual MBS data delivery comprises: transmitting, to a multicast/broadcast-session management function (MB-SMF), a session context update request to establish multicast MBS session transmission between a multicast/broadcast user plane function (MB-UPF) and a UPF; receiving, from the MB-SMF, a session context update response including multicast downlink (DL) tunnel information; and performing modification procedure for an N4 session based on the session context update response.
3. The method of claim 2, wherein the session context update request includes MBS session identifier (ID) and DL tunnel information of the UPF.
4. The method of claim 2, wherein the session context update response includes tunnel endpoint information of the MB-UPF.
5. The method of claim 1, further comprising: transmitting, to the target NG-RAN through the AMF, a response message for the first message.
6. The method of claim 1, wherein the first message is a protocol data unit (PDU) session update SM context request message including the N2 SM information, and wherein the N2 SM information is included in a N2 path switch request message transmitted by the target NG-RAN to the AMF.
7. A session management function (SMF) for Xn handover of a multicast/broadcast service (MBS) in a mobile communication system, the SMF comprising: a network interface configured to communicate with network functions (NFs) in the mobile communication system; a memory configured to store information; and at least one processor configured to control to: receive, from a target next generation radio access network (NG-RAN) node through an access and mobility management function (AMF) using the network interface, a first message including information on whether the target NG-RAN node supports the MBS, determine an individual delivery method for MBS data, in case that the target NG-RAN node does not support the MBS based on the information received from the target NG-RAN node, and perform an establishment of individual

MBS data delivery, wherein the information on whether the target NG-RAN node supports the MBS is N2 session management (SM) information received from the target NG-RAN node to the SMF via the AMF.

8. The SMF of claim 7, wherein at least one processor is further configured to control to: transmit, to a multicast/broadcast-session management function (MB-SMF), a session context update request to establish multicast MBS session transmission between a multicast/broadcast user plane function (MB-UPF) and a UPF; receive, from the MB-SMF, a session context update response including multicast downlink (DL) tunnel information; and modify with the UPF a N4 sessions based on the session context update response.

9. The SMF of claim 8, wherein the session context update request includes MBS session identifier (ID) and DL tunnel information of the UPF.

10. The SMF of claim 8, wherein the session context update response includes tunnel endpoint information of the MB-UPF.

11. The SMF of claim 7, wherein at least one processor is further configured to control to transmit, to the target NG-RAN through the AMF, a response message for the first message.

12. The SMF of claim 7, wherein the first message is a protocol data unit (PDU) session update SM context request message including the N2 SM information, and wherein the N2 SM information is included in a N2 path switch request message transmitted by the target NG-RAN to the AMF.
